



## DPUDX8Joystick

Provides an interface to a DirectX 8.0 compatible joystick.

### 1. Dynamic parameters:

|                       |                                                                                                        |
|-----------------------|--------------------------------------------------------------------------------------------------------|
| <i>upThreshold</i>    | Threshold from which <i>y</i> values are interpreted as <i>up</i> . Default: 0.2<br>Default value: 1 s |
| <i>downThreshold</i>  | Threshold from which <i>y</i> values are interpreted as <i>down</i> . Default: -0.2                    |
| <i>leftThreshold</i>  | Threshold from which <i>x</i> values are interpreted as <i>left</i> . Default: -0.2                    |
| <i>rightThreshold</i> | Threshold from which <i>x</i> values are interpreted as <i>right</i> . Default: 0.2                    |

### 2. Static parameters:

|                   |                                                                                                                                     |
|-------------------|-------------------------------------------------------------------------------------------------------------------------------------|
| <i>JoystickID</i> | ID of the joystick. 1 is the first connected joystick, 2 the second, etc.<br>Default: 1                                             |
| <i>NoError</i>    | If set to <i>true</i> , the simulation will not be stopped if an error occurs while accessing the joystick.<br>Default: <i>true</i> |

### 3. Outputs:

|                       |                                                                                                                    |
|-----------------------|--------------------------------------------------------------------------------------------------------------------|
| <i>x</i>              | Value of the joystick's X axis (normally corresponds to left/right movement, between -1 (left) and +1 (right))     |
| <i>y</i>              | Value of the joystick's Y axis (normally corresponds to back/forward movement, between -1 (back) and +1 (forward)) |
| <i>z</i>              | Value of the joystick's Z axis (often corresponds to the „throttle“ control, between -1 (low) and +1 (high))       |
| <i>rotx</i>           | Value of the joystick's rotation around the X axis (range: -1 to 1)                                                |
| <i>roty</i>           | Value of the joystick's rotation around the Y axis (range: -1 to 1)                                                |
| <i>rotz</i>           | Value of the joystick's rotation around the Z axis (range: -1 to 1)                                                |
| <i>up</i>             | Boolean output, which is set to true when <i>x</i> > <i>upThreshold</i> .                                          |
| <i>down</i>           | Boolean output, which is set to true when <i>x</i> < <i>downThreshold</i> .                                        |
| <i>left</i>           | Boolean output, which is set to true when <i>y</i> < <i>leftThreshold</i> .                                        |
| <i>right</i>          | Boolean output, which is set to true when <i>y</i> > <i>rightThreshold</i> .                                       |
| <i>Button[1...30]</i> | State of the buttons (0: unpressed, 1: pressed)                                                                    |
| <i>Slider[1...2]</i>  | Value of the sliders (range is not standardized)                                                                   |
| <i>Pov[1...4]</i>     | Position of the coolie hats („point of view hats“).<br>Unit: 1/100 degree                                          |
| <i>Pov[1...4]N</i>    | Boolean output, which is set to true when this coolie hat points north.                                            |



|                           |                                                                                     |
|---------------------------|-------------------------------------------------------------------------------------|
| <code>Pov[1...4]NW</code> | Boolean output, which is set to <i>true</i> when this coolie hat points north-west. |
| <code>Pov[1...4]W</code>  | Boolean output, which is set to <i>true</i> when this coolie hat points west.       |
| <code>Pov[1...4]SW</code> | Boolean output, which is set to <i>true</i> when this coolie hat points south-west. |
| <code>Pov[1...4]S</code>  | Boolean output, which is set to <i>true</i> when this coolie hat points south.      |
| <code>Pov[1...4]SO</code> | Boolean output, which is set to <i>true</i> when this coolie hat points south-east. |
| <code>Pov[1...4]O</code>  | Boolean output, which is set to <i>true</i> when this coolie hat points east.       |
| <code>Pov[1...4]NO</code> | Boolean output, which is set to <i>true</i> when this coolie hat points north-east. |

#### 4. Functional principle:

The DPUDX8Joystick represents an interface to any joystick that is supported by Microsoft DirectX 8.0. The axis range is standardized between -1 and +1. It is necessary to install Microsoft DirectX 8.0 or higher before operating the DPU.

If the connected joystick doesn't have so many axes, buttons, coolie hats or sliders, the corresponding DPU outputs are set to zero.

##### Note:

This DPU can also be used to interface with steering wheels designed for racing games.

However, the DPUDX8RacingWheel is optimized for steering wheels, and especially supports force feedback.

#### 5. Example:

```
DPUDX8Joystick ~Joy
{
    Computer = {MOCKUP};
    Index = 5;
    JoystickID = 1;
};
DPUJoystickConsole ~Mockup
{
    Computer = {MOCKUP};
    Index = 6;
    JoystickAMin = 0.0;
    JoystickAMax = 1.0;
    JoystickBMin = 0.0;
    JoystickBMax = -1.0;
};
```



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version Version 7.1 documentation

Connections =

```
{  
    ~Joy.x          -> ~Mockup.JoystickX_In,  
    ~Joy.y          -> ~Mockup.JoystickA_In,  
    ~Joy.y          -> ~Mockup.JoystickB_In,  
    ~Joy.Button1    -> ~Mockup.GearUp_In,  
    ~Joy.Button2    -> ~Mockup.GearDown_In,  
  
    ~Mockup.AcceleratorPedal -> Mockup.AcceleratorPedal,  
    ~Mockup.BrakePedal      -> Mockup.BrakePedal,  
    ~Mockup.SteeringWheel   -> Mockup.SteeringWheel,  
    ~Mockup.Gear            -> Mockup.Gear  
};
```